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# Brain Tumor Segmentation Comparitive Study

A Thesis Submitted in Partial Fulfillment of Mini Porject

By

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## Abstract

Brain tumors are dangerous and need to be detected early for better treatment. Doctors use MRI scans to find tumors, but manual detection takes time and can have errors. That's why machine learning and deep learning methods help in automatic tumor detection. This thesis compares four existing methods for brain tumor detection: CNN-based models, Neural Networks with Active Contour, traditional segmentation techniques, and a CNN-based ML approach. The study highlights their advantages, limitations, and accuracy levels. The findings show that deep learning models (CNNs) work better than older methods but still have some problems, like high computation cost and errors in detecting tumor edges. To solve these issues, this thesis proposes a Hybrid CNN-Transformer Model. This model uses CNN for feature extraction and Transformers to improve accuracy by capturing more details. It helps in better boundary detection, reducing errors, and improving accuracy. This research gives insights into which method works best for brain tumor detection and suggests a new approach for better accuracy. Future work includes using 3D MRI scans and combining multiple imaging techniques for even better results.

**Keywords:** Brain Tumor Detection, MRI, Deep Learning, CNN, Transformers, Medical Imaging.

## 1 Introduction

A **brain tumor** is an abnormal cell growth in the brain that can be either **benign (non-cancerous)** or **malignant (cancerous)**. Early detection is crucial for better treatment outcomes. **MRI scans** are the primary method for diagnosing brain tumors, but manual analysis is **time-consuming and error-prone**. Deep learning models, especially **Convolutional Neural Networks (CNNs)**, have shown high accuracy in tumor segmentation and classification.

This study compares **four existing models** used in brain tumor detection and proposes a **Hybrid CNN-Transformer Model** to address accuracy and efficiency challenges.

## 2 Literature Review

### Key Findings:

- CNN-based models provide **better accuracy** than traditional methods.

| Method                          | Accuracy                | Efficiency                     |
|---------------------------------|-------------------------|--------------------------------|
| U-Net CNN (Paper 1)             | High (Dice Score: 0.88) | Computationally expensive      |
| Active Contour Model (Paper 2)  | Moderate                | Faster than CNN                |
| Comparative Study (Paper 3)     | Varies by method        | Traditional methods are slower |
| CNN + Hough Transform (Paper 4) | 91% accuracy            | More efficient than U-Net      |

Table 1: Comparison of Existing Methods

- Hybrid approaches improve efficiency but depend on **accurate boundary detection**.
- A **Hybrid CNN-Transformer Model** could enhance segmentation by combining CNN’s feature extraction with Transformer’s long-range dependency capture.

## 3 Proposed System: Hybrid CNN-Transformer Model

### 3.1 Objective

- Improve **brain tumor segmentation accuracy**.
- Reduce **false positives & false negatives**.
- Enhance **tumor boundary detection** using Transformers.

### 3.2 Methodology

- **Dataset:** BraTS (Brain Tumor Segmentation)
- **Preprocessing:** Skull stripping, normalization, contrast enhancement
- **Segmentation:** CNN for feature extraction + Transformer for global context understanding
- **Classification:** Fully Connected (FC) layers in CNN for tumor classification
- **Performance Metrics:** Dice Score, Sensitivity, Specificity, Accuracy

### 3.3 Advantages of Proposed System

- Better tumor **boundary detection** than CNN-only models.
- Reduces **segmentation errors & misclassifications**.
- Works on **limited datasets** more efficiently.
- Suitable for **real-world medical applications**.

## 4 Conclusion & Future Scope

Brain tumor detection is a critical medical challenge, and deep learning models have significantly improved accuracy. This study compared four existing models and highlighted their limitations. To address these, a **Hybrid CNN-Transformer Model** is proposed, aiming for better segmentation accuracy and real-world applicability.

#### Future Work:

- Extend the model for **3D MRI segmentation**.
- Optimize for **low-cost real-time implementation**.
- Improve interpretability using **attention mechanisms**.

With the proposed improvements, brain tumor detection can become more **accurate, efficient, and accessible** for clinical use.

## 5 References

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